

FEATURE ARTICLE

Oral Herbal Medicines for Psoriasis: A Review of Clinical Studies

Brian H. May¹, Anthony L. ZHANG¹, Wenyu ZHOU¹, LU Chuan-jian (卢传坚)²,
Shiqiang DENG¹, and Charlie C.L. XUE^{1,2}



Prof. Charlie C.L. XUE

ABSTRACT Various forms of complementary and alternative medicine are used in psoriasis. Among these, herbal medicines are frequently used as systemic and/or topical interventions either as a replacement for or in conjunction with conventional methods. The benefit of such use is unclear. This review is to provide an up-to-date review and discussion of the clinical evidence for the main kinds of herbal therapies for psoriasis. Searches of the biomedical databases PubMed (including MEDLINE), EMBASE and CINAHL were conducted in December 2011 which identified 32 clinical studies, all published in English. Twenty of these primarily tested topical herbal medicines and were thus excluded. The 12 studies that evaluated systemic use of herbal medicines were included in the review. Four were case series studies and the other 8 were controlled trials. In terms of interventions, 4 studies tested the systemic use of plant oils combined with marine oils and 8 studies tested multi-ingredient

herbal formulations. The clinical evidence for plant and animal derived fatty acids is inconclusive and any benefit appears to be small. For the multi-herb formulations, benefits of oral herbal medicines were shown in several studies, however, a number of these studies are not controlled trials, a diversity of interventions are tested and there are methodological issues in the controlled studies. In conclusion, there is promising evidence in a number of the studies of multi-herb formulations. However, well-designed, adequately powered studies with proper control interventions are needed to further determine the benefits of these formulations. In addition, syndrome differentiation should be incorporated into trial design to ensure effective translation of findings from these studies into Chinese medicine clinical practice.

KEYWORDS psoriasis, herbal medicine, complementary medicine, traditional medicine, clinical study, review

In a recent review of complementary and alternative medicine (CAM) for psoriasis, Smith, et al⁽¹⁾ discussed outcomes of randomised controlled trials (RCTs) of a wide range of topical and systemic therapies including vitamin and mineral supplements; fish and plant oils; herbal medicines; bathing in the Dead Sea and other kinds of bath or spa therapy; and mind-body therapies such as psychotherapy, meditation, hypnosis, stress reduction and bio-feedback. These CAM therapies were used alone or in combination with conventional therapies such as pharmacotherapy and ultra-violet (UV) irradiation and were compared with active treatments or placebo. The reviewers concluded that the evidence was conflicting, based on only a few studies, and there were methodological deficiencies in many of the studies. A subsequent review by Reuter, et al⁽²⁾ that focussed on clinical and experimental studies of herbal medicine concluded that the most promising herbal medicines for psoriasis were capsaicin and Mahonia aquifolium used topically. There is a lack of systematic

evaluation on the benefits of oral use of herbal medicine for psoriasis.

This review examines the clinical research literature published in English on the treatment of psoriasis using herbal medicines (HMs) administered systemically. Studies of phototherapy, diet and vitamin use, bath and spa therapies, homeopathy, and psychological interventions were excluded. HM was broadly defined to include natural

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1. Traditional & Complementary Medicine Research Program, Health Innovations Research Institute; WHO Collaborating Centre for Traditional Medicine; School of Health Sciences, RMIT University, Bundoora, VIC 3083, Australia; 2. Guangdong Provincial Academy of Chinese Medical Sciences and Guangdong Provincial Hospital of Chinese Medicine, Guangzhou (510405), China
Correspondence to: Prof. Charlie C.L. XUE, Tel: 613- 99257360, E-mail: Charlie.xue@rmit.edu.au
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products of plant, animal and mineral origin but dietary regimens and fish oil supplements were excluded except when combined with HMs. HMs could be used singly or in formulations, as extracts, or could be combined with topical HMs or non-herbal ingredients. In order to provide a comprehensive view of the state of the clinical evidence, a wide range of study designs have been included.

METHODS

Searches were conducted in December 2011 of the major biomedical databases PubMed (including MEDLINE), EMBASE and CINAHL using a search strategy designed to capture a broad range of clinical studies and reviews of CAM for psoriasis. No limits were set for publication year and all articles were published in English. Additional citations were obtained by scrutinising review articles. Controlled trials and case series studies have been included but not individual case reports. Each article was examined by two authors, data were extracted to tables and the articles classified according to the type of intervention. Since this review includes studies other than RCTs, methodological issues are discussed in relation to individual studies.

RESULTS

Citations identified in the searches were examined and 111 journal articles retrieved including 63 clinical studies and 28 reviews. Following the exclusion process, 32 clinical studies published in English were identified. Of these, the 12 that focused on the systemic use of HM were included in the review.

The 12 included studies were categorised as follows: 4 studies of plant plus marine oils used orally, and 8 studies of multi-ingredient herbal formulations used systemically, 7 of which were for Chinese HMs. The characteristics of the studies of systemic use of HMs are summarised in Table 1. Four studies were case series and the other 8 were controlled trials. All 4 studies of plant plus marine oils were RCTs that compared the HM with placebo. For the multi-ingredient HMs, 4 were case series studies and the others used comparisons with placebo and/or active controls.

DISCUSSION

The studies were classified according to the main test intervention with the 4 studies on the systemic use plant oils combined with marine oils being discussed first, followed by the 8 studies on multi-ingredient herbal

formulations.

Plant Plus Marine Oils (4 Studies)

Fatty-acid containing oils have been used in psoriasis for a number of years but the clinical evidence is mixed. A combination of Evening primrose oil (from *Oenothera biennis* and related species), fish oil, and vitamin E which is sold under the trade name "Efamol Marine" has been investigated in 3 studies conducted in Britain. Strong and Hamill conducted a double blind RCT in 51 in-patients over 7 months but found no significant difference between active and placebo groups.⁽³⁾ Oliwiecki and Burton employed a double-blind placebo controlled design in 37 patients with chronic stable plaque psoriasis over 24 weeks. Patients could also apply emollients as required and used 1% hydrocortisone ointment twice daily. No differences were found in clinical symptoms or plaque thickness.⁽⁴⁾ A similar study of 38 patients by Veale, et al⁽⁵⁾ found that the same product produced no significant change in clinical symptoms after 9 months of use, however, serological measurements suggested an anti-inflammatory effect.

Another commercially available product composed of herbal and marine ingredients, called "HESA-A", was compared with placebo in 28 patients with chronic plaque psoriasis in Iran. It was reported to have improved clinical symptoms with no adverse reactions.⁽⁶⁾ This medicine has also been reported to have benefits in other diseases and a number of experimental studies have been published but the ingredients of HESA-A are not specified and all identified studies were conducted by the same research group.^(7,8)

Eicosapentaenoic acid (EPA), docosahexaenoic acid (DHA) and other fatty acids found in plant and animal oils have shown anti-inflammatory effects in experimental and clinical studies of other disorders.⁽⁹⁾ In psoriasis, a clinical study of oral fish oil found a small benefit in reducing topical steroid use.⁽¹⁰⁾ A marked improvement has been reported for intravenous fish oil,⁽¹¹⁾ and oral seal oil has also been reported to improve symptoms.⁽¹²⁾ A review on topical EPA use concluded it has a potential role either alone or in combination with other agents.⁽¹³⁾ Interestingly, a recent study found a synergistic anti-inflammatory effect for the combination of curcumin and EPA.⁽¹⁴⁾ Efamol Marine and related products are still available as sources of fatty acids (<http://www.efamol.com>), however, recent clinical studies on fatty acids from plant oils for psoriasis could not be located.

Table 1. Characteristics of Studies of Systemic Herbal Medicine Interventions

First author, year, location duration	Study type Comparison Disorder	Participants Treatment group Control group	Test intervention, dose Ingredients	Control intervention, dose Ingredients	Outcome measures	Dropouts AEs
Ahmadi, A. 2008, Iran, 6 months	RCT, double blind, 2 groups HM vs. placebo Chronic plaque-type psoriasis	28 (11 M), mean age 31.07 ± 4.5 yrs T: 14 (5 M) C: 12 (6 M)	Oral HESA-A tablets (25 mg/kg), twice a day Ingredients not specified	Identical placebo tablet	Disease severity (6 point scale)	No dropouts No AEs
Chang, S. 2006, China, 8 weeks	RCT, 2 groups HM1 vs. HM2 psoriasis vulgaris of blood heat type	120 T: 60 (34 M), mean age 35.47 ± 12.5, range 18–65 yrs; C: 60 (31 M), mean age 36.40 ± 11.32, range 21–65 yrs	Yinxieping Granule, 4.5 g, twice a day Radix Rehmanniae, Radix Angelicae Formosanae, Powder of Carapax Erethochelydis, Radix Paeoniae Rubra, Calculus Bovis Artificial, Herba Schizonepetae Tenuifoliae	Xiaoyin Tablet, 7 tablets, thrice a day Radix Rehmanniae, Paeonia suffruticosa, aconia veitchii, Angelica sinensis, Sophora flavescens, Lonicera japonica, Scrophularia ningpoensis, Arctium lappa, Cyptotypana atrata, Dictamnus dasycarpus, Saposhnikovia divaricata, etc.	Therapeutic effectiveness, scaling, erythema, pruritus	No dropouts No AEs or abnormalities in medical test Results were identified
Ho, S.G. 2009, China, Hong Kong 6 months	RCT, double blind, 3 groups HM vs. methotrexate (MTX) vs. placebo Note: MTX group not double blind Moderate to severe plaque psoriasis	61 HM: 21 (14 M), mean age 48.52, range 25–80 yrs; MTX: 20, (18 M), mean age 38.45, range 21–68 yrs; Placebo: 20, (18 M), mean age 43.45, range 27–61 yrs	Wentong Huayu Capsule (dose not specified) Herba ephedrae 6 g, Radix aconiti lateralis preparata 10 g, Semen sinapis 10 g, Cortex cinnamomi 3 g, Rhizoma zingiberis 3 g, Cornu cervi degelatinatum 15 g, Radix rehmanniae preparata 10 g, Rhizoma smilacis glabrae 60 g, Cortex dictamni 30 g, Rhizoma imperatae 30 g, Radix salviae miltiorrhizae 15 g, Caulis spatholobi 30 g, Radix arnebiae 30 g, Flos sophorae 30 g, Radix glycyrrhizae 6 g, Indigo naturalis 6 g	Methotrexate: 2.5–5 mg then increasing up to 30 mg/week + Folic acid 5 mg daily; or placebo capsule identical to HM capsule (dose not specified)	PASI; Physician's global assessment (PGA); Psoriasis disability index (PDI)	11 dropouts AEs: MTX group 65%, HM group 48%; placebo group 30%
Liu, H. 2004, China, 60 days	Case series HM vs. healthy group Chronic plaque psoriasis of blood stagnation due to qi deficiency type	HM: 40 (26 M), mean age 60, range 60–69 yrs; Healthy control: 31 (20 M), mean age 64, range 30–68 yrs	Oral HM, twice a day plus topical 10% boric acid ointment HM: Radix Astragali 60 g, Radix Codonopsis 30 g, Radix Salviae Miltiorrhizae 30 g, Radix Paeoniae Rubra 30 g, Rhizoma Chuansong 10 g, Pheretima 10 g, Radix Achyranthis Bidentatae 15 g, Radix Arnebiae seu Lithospermi 15 g, Radix Glycyrrhizae 6 g. Addition of Radix Scrophulariae for dry mouth/thirst, Cortex Dictamni for pruritus	No treatment	Improvement rate; CD3, CD4, CD8 T lymphocyte levels, CD4/CD8 ratio; hemorheology	No dropouts AEs not mentioned
Lone, A.H. 2011, India, 8 weeks	RCT, single-blind, 2 groups HM1 + HM2 vs. placebo 1 + placebo 2 Plaque-type psoriasis	30 (21 M), 11–60 yrs T: 20 C: 10	Oral Majoon Ushba, 5 g, twice a day, plus topical Roghane Hindi, twice a day Majoon Ushba: Smilax officinalis 40 g, Cassia angustifolia 80 g, Pterocarpus santalinus 60 g, Santalum album 60 g, Smilax china 60 g, Rosa damascena 60 g, Cinnamomum zeylanicum 40 g, Piper cubeba 40 g, Borage officinalis 40 g, Cuscuta reflexa 40 g, Polypodium vulgare 40 g, Terminalia bellerica fruit 20 g, Nardostachys jatamansi 20 g, Terminalia chebula unripe 15 g, Terminalia chebula half ripe coat 10 g, powdered and mixed with water and 2 kg white sugar	Oral placebo (wheat flour), 5 g twice a day plus topical placebo (coconut oil), twice a day Topical test intervention Roghane Hindi: Azadirachta indica 250 g, Copper sulphate 10 g, Monoxide of lead 10 g, Tamarix gallica 10 g, Terminalia chebula 10 g, Curcuma longa 10 g, and Mustard oil (250 mL)	PASI score; itching, scaling, erythema, liver and kidney function	No dropouts No abnormality in liver or kidney function
Oliwiecki, S. 1994, UK, 4 weeks run-in, 24 weeks treatment	RCT, double blind, 2 groups HM vs. placebo Chronic stable plaque psoriasis	37, age range 16–70 yrs T: not specified C: not specified	Efamol Marine, 12 capsules/day, plus emollients & Efcortelan (1% hydrocortisone) as required 430 mg Evening primrose (Oenothera biennis) oil + 107 mg fish oil + 10 mg Vit E	Placebo, 12 capsules (500 mg liquid paraffin)/day, plus emollients & Efcortelan (1% hydrocortisone) as required	Symptom severity, plaque thickness	No information provided

(To Be Continued)

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First author, year, location duration	Study type Comparison Disorder	Participants Treatment group Control group	Test intervention, dose Ingredients	Control intervention, dose Ingredients	Outcome measures	Dropouts AEs
Strong, A.M.M. 1993, UK 7 months	RCT, double blind, 2 groups HM vs. placebo Chronic stable plaque psoriasis	51 (22M) T: 26 C: 25	Etamol Marine, 6 capsules (500 mg/capsule), twice a day, plus 2% salicylic acid ointment	Placebo, 6 capsules (600 mg liquid paraffin), twice a day, plus 2% salicylic acid ointment	Symptom severity (VAS), skin area, itch, plasma dihomogammalinolenic acid (DGLA), blood tests	24 dropouts: psoriasis deterioration (T:5, C:3), others due to non-compliance AE: nausea (T:1)
Tang, Y.Q. 2005, Singapore 10 months	Case series Moderate to severe chronic plaque psoriasis	15 enrolled 11 completed (10 M), age range 18–65 yrs	Oral Herose, 4 capsules (450 mg/capsule), thrice a day <i>Rhizoma Zingiberis</i> 454 mg, <i>Radix Salviae Miltiorrhizae</i> 589 mg, <i>Radix Astragal</i> 331 mg, <i>Ramulus Cinnamomi</i> 317 mg, <i>Radix Paeoniae alba</i> 165 mg, <i>Radix Codonopsis pilosula</i> 67 mg, <i>Semen Cois</i> 570 mg	No control Patients also used topical Emollient	PASI	4 dropouts, various reasons AEs: headache (1), nausea (1)
Tang, Y.Q. 2008, Singapore [letter]	Case series Psoriasis (1 guttate type, 1 plaque type)	3 1 female 12 yrs, 2 males 38 & 52 yrs	Oral Herose, 1,440 mg, thrice a day <i>Rhizoma Zingiberis</i> 429 mg, <i>Radix Salviae Miltiorrhizae</i> 556 mg, <i>Radix Astragal</i> 313 mg, <i>Ramulus Cinnamomi</i> 299 mg, <i>Radix Paeoniae alba</i> 165 mg, <i>Radix Codonopsis pilosula</i> 63 mg, <i>Semen Cois</i> 538 mg	No control	PASI	No AEs associated with herose
Veale, D.J. 1994 Britain, 9 months treatment, 3 months follow-up	RCT, double blind, 2 groups HM + NSAID vs. placebo + NSAID Chronic stable plaque psoriasis & inflammatory arthritis	38 T: 19 (7 M), mean age 40, range 25–58 yrs; C: 19 (7 M) mean age 40, range 18–76 yrs	Etamol Marine, 12 capsules/day (480 mg GLA, 240 mg EPA, 132 mg DHA/day) Evening primrose (<i>Oenothera biennis</i>) oil + fish oil + Vit E	Placebo, 12 capsules/day (paraffin + Vit E) Patients in both groups continued NSAID medication with progressive reduction if possible	VAS (disease severity), NSAID use, blood tests (hemoglobin, ESR, CRP, immunoglobulins, urea, electrolytes and liver enzymes)	6 dropouts (T:6, C:0) AEs: diarrhea (2), migraine exacerbation (1), symptoms worse (2)
Wang, G. 2004, China, 1 month [letter]	Case series, 3 groups 3 different HM administered orally according to stage of disease: HM 1: developing stage; HM 2: stable stage; HM 3: remission stage HM 1 or HM 2 or HM 3 psoriasis	72 No further details	HM 1: Buffalo horn, Spreading healyotis herb, Chinese corktree bark, and Snaking amebia root; or HM 2: <i>Radix Salviae Miltiorrhizae</i> , Zedocary, Oyster shell, <i>Scrophularia</i> ; daily HM 3: <i>Rehmannia</i> dried rhizome, <i>Tuber leeceflower</i> root, Chinese angelica, <i>Suberect spatholobus</i> stem; daily	No control All patients also used topical urea cream	Reduction in PASI scores	Dropouts not mentioned AEs: slight nausea (5)
Zhang, L.X. 2009, China, 8 weeks	RCT, 2 groups, HM + acitretin vs. HM Psoriasis of blood-heat type	80 Group A: 39 (37 M), mean age 42.6, range 22–67 yrs; Group B: 41 (37 M), mean age 43.1, range 24–70 yrs	Group A: oral HM decoction, twice a day, plus Qingkailing Injection 40 mL/day, IV, plus acitretin capsule 20–30 mg/day up to 40 mg/day, plus topical 5% boric acid ointment Qingkailing Injection: <i>Buffalo horn</i> , <i>Scutellaria</i> , <i>Honeysuckle</i> , <i>Cape jasmine</i>	Group B: oral HM decoction, twice a day, plus Qingkailing Injection 40 mL/day IV, plus topical 5% boric acid ointment Oral HM: <i>Dandelion</i> 10 g, <i>Forsythia</i> fruit 12 g, <i>Isatis</i> root 30 g, <i>Isatis</i> leaf 15 g, <i>Imperata</i> rhizome 30 g, <i>Honeysuckle</i> flower 15 g, <i>Prunella</i> spike 15 g, <i>Moutan</i> bark 15 g, <i>Red and white peony</i> each 15 g, <i>Rehmannia</i> root 15 g, <i>Figwort</i> root 15 g, taken with <i>Antelope horn powder</i> 0.3 g (modified according to symptoms)	Effectiveness rate, PASI	One dropout due to a pulmonary tumor AEs: mild or moderate symptoms reported

Notes: M: males; T: treatment group; C: control group; AE: adverse event; NSAID: non-steroidal anti-inflammatory drugs; ESR: erythrocyte sedimentation rate; CRP: C-reactive protein; EPA: eicosapentaenoic acid; GLA: gamma-linoleic acid; DHA: docosahexaenoic acid

Multi-Ingredient Systemic Herbal Formulations (8 Studies)

Eight reports on the systemic administration of multi-ingredient herbal formulations, 7 of which used Chinese HMs, were included in this review. These were diverse in both study design and in the HMs used with 4 studies reporting on series of cases and 4 employing various controlled comparisons (Table 1).

Liu and Tan treated a series of 40 elderly chronic plaque psoriasis patients aged 60 to 69 years who satisfied the diagnostic criteria for blood stagnation due to qi deficiency. Treatment involved a multi-herb decoction taken for 60 days plus the topical use of 10% boric acid ointment. In the 2 months prior to the treatment, patients did not take steroids or anti-psoriatic medications. Efficacy was scored on a 4-point scale with a 70% decrease in lesions being classified as "markedly effective". Serological tests were conducted before and after treatment to detect changes in CD4/CD8 T-lymphocyte ratio and blood viscosity parameters as surrogate measures of immune system function and blood stagnation. These were compared with results from 31 healthy adults aged 60–68 years who were found to have a higher CD4/CD8 ratio compared to the psoriasis patients at baseline. After treatment, 60% of patients experienced almost complete clearance of lesions, the CD4/CD8 ratio significantly increased but remained lower than in the healthy control, and measures of blood viscosity improved. There was no report of adverse events and it appears that all patients completed the study.⁽¹⁵⁾ These results suggest the HM improved immune function and reduced blood viscosity. Also, some of the herbs included in the decoction have been shown to have these effects in experimental studies.^(16,17) Based on these findings, the intervention should be further tested through a well-designed placebo controlled study to confirm the clinical efficacy observed in this case series study.

A letter from Wang, et al⁽¹⁸⁾ reported a study of three HM formulas administered to 72 psoriasis patients according to whether the disease was in the developing stage, stable stage or remission stage. Treatment continued for 1 month, urea cream was used topically but patients did not take any other systemic drugs. Reduction in psoriasis area and severity index (PASI) scores of 75% or greater were reported in 26 patients and there were no severe adverse events. However, this report was brief and there was no mention of a control group, the Chinese medicine diagnosis or other methodological aspects. Another uncontrolled study conducted in Singapore of 15 patients

with chronic plaque psoriasis reported that a multi-herb Chinese HM formula called "Herose" improved PASI scores over 10 months. Progress followed a wave-like pattern of improvement followed by rebound then further improvement with 6 of the 11 patients who completed the study achieving PASI-75 or more.⁽¹⁹⁾ In a subsequent letter that included photographs, Tang⁽²⁰⁾ reported on 3 successful cases also treated with Herose. One 12-year-old girl achieved total clearance in 10 weeks and two men who had proven to be resistant to conventional therapy achieved PASI-75 and PASI-50 after 9 and 11 months, respectively.⁽²⁰⁾ The type of psoriasis according to Chinese medicine differentiation was not specified. The above three case series studies all reported promising improvements in psoriasis symptoms but without the use of a control group, the results of these studies can only be considered preliminary.

Chang, et al⁽²¹⁾ reported on a comparison between two Chinese herbal formulas, Yin Xie Ping Granule (银屑平颗粒) and Xiaoyin Tablet (消银片) in out-patients diagnosed with psoriasis vulgaris of blood heat type. Each formulation was administered to 60 patients who were not taking another anti-psoriatic medicine over 8 weeks with a follow-up at 1 year. Yin Xie Ping Granule was the test intervention and the patients in this group were reported to have had statistically significant improvements in scaling compared to the Xiaoyin Tablet group but the overall therapeutic effectiveness was similar for the two formulations. Data for the follow-up were not reported but it was noted that the 17 patients classified as "cured" in the treatment group had not relapsed.⁽²¹⁾ While this study did involve a direct comparison between two groups, one of which (Xiaoyin Tablet) is frequently used for psoriasis in China, it was not blinded and a true control group was not used. Therefore, even though one HM was statistically superior to the other, this study cannot demonstrate that either HM was actually superior to placebo.

An RCT of 80 psoriasis patients of the blood heat type, Zhang, et al⁽²²⁾ compared a Chinese herbal therapy consisting of a multi-herb decoction plus Qingkailing Injection (清开灵注射液) with the same therapy plus acetretn capsule. Both groups were treated for 8 weeks and efficacy was evaluated based on change in PASI scores. The total effectiveness rate was higher in the HM plus acetretn group but the differences between groups in PASI scores were not significant. A range of adverse events were reported in both groups which were managed by medication adjustments but these were fewer in the HM only group. The authors concluded that Chinese HM plus acetretn was

more effective than the HM alone and also relatively safe. However, the study was not blind, no placebo for acetretin was used and there was no direct comparison with acetretin alone, therefore, it is not possible to conclude that the addition of the HM conferred an additional benefit or vice-versa.

Ho, et al⁽²³⁾ undertook a direct comparison between a Chinese herbal formulation Wentong Huayu Capsule (溫通化療膠囊) and placebo using a double-blind RCT over 6 months which also included an unblinded comparison with methotrexate. The 61 patients ceased other medications except fluocinolone 0.0125% cream and aqueous cream. Significant differences were found in PASI scores at 2, 4 and 6 months for the methotrexate group compared to both HM and placebo which showed no differences between groups. Similarly, Physician's global assessment (PGA) scores were only significant for the methotrexate group. The Psoriasis disability index (PDI) declined for all 3 groups with no significant inter-group difference at 6 months and the effect of the HM was not statistically different to placebo on any outcome measure. The placebo was surprisingly effective in this study resulting in a 32% reduction in PASI over 6 months compared with 73.9% reduction in the methotrexate group. Adverse events were reported by a higher percentage of patients in the methotrexate group (65%) compared to HM (48%) or placebo (30%) but the dropout rates were higher in the HM and placebo groups with only 14 and 17 patients completing the study. The researchers noted that this HM formulation has a history of use in Hong Kong with satisfactory results but this was in decoction form rather than as the standardised capsule used in this study and the dosages may not have been comparable.⁽²³⁾ This was a well-designed study in terms of methodology but Chinese medicine differential diagnosis appears not to have been used in selecting subjects for the HM and other groups, so it is unclear whether the test HM was appropriate for all patients.

A recent single-blind, properly randomised study in India compared the combination of an oral plus a topical traditional Unani multi-ingredient formulation with an oral plus a topical placebo for 8 weeks. The test group comprised 20 patients with 10 in the placebo group. Patients did not take anti-psoriasis medications in the preceding 2 months or during the study. The oral medication was Majoon Ushba and the topical medicine was the oil Roghane Hindi. A significant difference was found between groups with no change in the placebo

group (PASI 6.8 average) and a decline in the Unani medicine group to an average of PASI 3.25. Liver and kidney function test returned no abnormal results. The effects of the individual HMs were discussed according to traditional theory and modern research.⁽²⁴⁾ Although the study was described as single-blind, it was not clear whether the placebos were identical to the test medications and blinding success was not reported.

Positive effects were reported in 7 of the 8 reports for multi-ingredient HMs. In one of the 2 RCTs that compared HM with placebo, a considerable placebo effect was found and the HM was found to have no additional benefit.⁽²³⁾ In the other, the placebo had no effect on symptoms and the combination of systemic plus topical herbal interventions were found to be effective.⁽²⁴⁾ However, the treatment groups in both studies were small ($n=21$ and $n=20$, respectively) and the HM interventions were quite different.

For the Chinese HMs, 2 case series studies referred to the same HM^(19,20) but the other studies all tested different HMs. Although differential diagnosis is typically used in Chinese medicine practice to determine the appropriate HM, only 3 studies employed this strategy.^(15,21,22) Two of these focussed on psoriasis of the blood heat type^(21,22) and one specifically investigated psoriasis in older people who had psoriasis characterised by blood stasis.⁽¹⁵⁾ A survey has found that blood heat is the most common type, followed by the blood dryness and blood stasis types.⁽²⁵⁾ The former type was typical of recent cases and of the aggravation stage while the latter two types were more evident in longer term cases. Considering the tendency for psoriasis to demonstrate a pattern of aggravation and remission, as noted by Tang,⁽¹⁹⁾ when conducting studies it is important to take both the syndrome differentiation and the stage of disease development into consideration.

Conclusions

The clinical evidence for plant and animal derived fatty acids is inconclusive and any benefit appears to be small. For the multi-herb formulations, there are a number of promising reports in the English language literature of benefit for various HMs used systemically but only a few RCTs have been published. These employed different interventions and the 2 placebo-controlled trials were based on relatively small samples. Therefore, based on the evidence reported in the English language literature, it is not possible to conclude that the reported effects were due to the treatment interventions alone.

Larger well-designed studies that compare the HMs with appropriate placebos and employ blinded assessments are needed. These should continue over a period of time sufficient to determine the real clinical effect of the commonly used HMs. In the case of studies of Chinese HMs, patients need to be treated according to their syndrome and treatment may require individual modification of the prescription. These present challenges in terms of study design, therefore, it may be prudent to commence with an open-label study of a well-defined group of patients using blinded assessments to refine the methodological aspects and provide data for power calculations prior to undertaking a controlled study.

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